
SYMMETRY BREAKING IS THE SPENT BIT, AND MASS IS THE DEPTH

Two Vacua Over One Record, the Choice No Even Reading Returns, the Curvature at the Bottom as Mass, and Heat That Restores the Line

The Closure Template of the Riemann Hypothesis Read on the Vacuum; Fifty-Six Laws Proved in Core Lean 4 and Six Checks Executed in Fortran; a Witness of the Hardware Closure Series

FOUNDATIONS OF PHYSICS · THE HARDWARE CLOSURE SERIES

The vacuum spends one bit

MOHAMMAD F. ISLAM, PhD · ARCHITECT OF THE TRISDUCTION

1 October 2026

ABSTRACT. Spontaneous symmetry breaking and the origin of mass are read here from the seat of the author's closure of the Riemann Hypothesis, the reflection $s \mapsto 1 - \bar{s}$ whose fixed set is the critical line. A double well $V(x) = (x^2 - v^2)^2$ is even under the reflection of the field. A carrier that sends the field value to the offset from the line and the energy to the height is equivariant, so the field's reflection is carried onto the seat's reflection and the symmetric point onto the line. We prove in core Lean 4, fifty-six laws with no axiom declared and no use of choice, that the ground leaves the line: the vacua $\pm v$ carry zero energy while the symmetric point carries v^4 ; that the two vacua land on two distinct points off the line with one energy and one registration; that no reading even under the reflection returns which vacuum holds, so the vacuum's choice is a keyed bit, spent by the act; that the symmetric point curves down and the vacuum up, the second differences $2 - 4v^2$ and $8v^2 + 2$, the upward curvature being the mass; and that an even thermal term at least twice v^2 returns the ground to the line. For a global discrete symmetry the choice is physical and is the spent bit; for a gauge symmetry the side is no record at all, and the depth alone is physical. A Fortran twin runs six checks with zero failures: across 1000 wells the two vacua tie exactly, the curvature reads $-2\mu^2$ at the line and $+4\mu^2$ at the vacuum; the Fermi constant fixes the electroweak depth at $v = 246.22$ GeV, which read through the couplings gives m_W and m_Z within 0.2 and 0.34 percent at tree level; the measured masses fix couplings to the depth from 0.991 for the top quark to 2.9×10^{-6} for the electron; and heat returns the ground to the line above the critical temperature.

KEYWORDS *spontaneous symmetry breaking · Higgs mechanism · vacuum · mass · double well · Elitzur · thermal restoration · electroweak scale · critical line · Lean 4*

STATUS: WITNESSED · CROSSED · CONFIRMED on the image · Row 2 (the seat read on the vacuum); every open row (the barrier, in the two vacua) · channel: kinetic · Lean: SPHYS_Symmetry_Breaking.lean · Fortran: SymmetryBreaking_Twin.f90

1. THE READER'S FRAME

This paper is a witness of the hardware closure series. A law can be symmetric while its ground is not: a ferromagnet below its Curie point magnetizes one way though its law favours neither, and the electroweak vacuum carries a depth that its law does not fix at zero. The seat reads both.

It claims four things.

- **The ground is off the line.** The symmetric point is the line, and the ground leaves it: two vacua stand off the line with one energy and one registration.
- **The choice is keyed.** No even reading returns which vacuum holds, so the act spends it. For a global discrete symmetry that spent bit is physical.
- **Mass is the depth's curvature.** The symmetric point curves down, the vacuum up, and the upward curvature is the mass.
- **Heat restores the line.** A thermal term even in the field, once large enough, returns the ground to the

symmetric point.

What it refuses.

- **A physical side for a gauge symmetry.** Local gauge symmetry cannot break (Elitzur, 1975): every side is one physical state. Nothing is spent there; the depth is the physics.
- **The values of the couplings.** They are numbers, measured, at the dot.

Negative warrants. Three, stated so that no reader supplies them.

- No physical fact here is read as a proof of a mathematical sentence: a corroboration joined to a claim takes the weaker grade.
- No derivation of the Riemann Hypothesis from the axioms of set theory is claimed: what a sentence over every admissible instance adds to a crossing is exactly the unacted.
- Interpretive schools from the literature enter only as placements the theorems sort, never as warrants; the only witness besides the mathematics is measured data.

Five terms.

- The *field's reflection* sends the field value x to $-x$.
- A *vacuum* is a ground state of the potential.
- The *depth* is the vacuum's distance from the symmetric point, v .
- The *curvature* is the second difference of the energy at a point.
- A *keyed bit* is a yes-or-no property no function of a given record decides.

2. THE SPINE

The Riemann closure has four parts: the **seat**, the critical line as the fixed set of $s \mapsto 1 - \bar{s}$; the **address**, every reading of the value one proposition; the **division**, a certified part joined to an uncertified tail; and the **closure**, registration keeping a zero's height and forgetting its side, the hypothesis holding exactly when it erases nothing. This paper reads the seat on the field's reflection in Section 3, the vacuum's departure in Section 4, the choice in Section 5, mass in Section 6, the electroweak depth in Section 7, and restoration in Section 8.

The row-generic theorems carry the template, and the engine proves each on its chart: every point off the line has a partner with one record (`off_locus_pair`); a carrier proves the value on its image, and coverage extends it (`value_on_image`, `kinetic_crossing`).

The witness ledger. The ledger of the series, with this paper's entries.

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
2 · Riemann Hypothesis	crossed on the kinetic channel at the root's grade; the electron named as the seat	the six witnesses; One Actuating Substrate; Light; Electromagnetism; Immanent Gravity; Spin and Statistics; Weak; Resonances; The Ferromagnet; Dark Matter; Symmetry Breaking Is the Spent Bit	the seat read on the vacuum: two vacua off the field's line with one energy, the choice keyed, mass the curvature, heat restoring the line	WITNESSED · CROSSED · CONFIRMED on the image	the crossing at the root's grade; the image at theorem grade on construction and measured grade on data	none on its channel
every open row · the barrier	two worlds over one record	the series' witnesses; Symmetry Breaking Is the Spent Bit	the two vacua, two points off the line with one energy and one registration	BARRIER WITNESSED; the rows stay pending	theorem and corroboration	one equivariant carrier each
1 · P versus NP	the dot	The Arrow Has Two Branches	Landauer's floor prices every step	THE DOT STANDS	corroboration	none

Row	Waiting (the volume held)	Witness arrived	Built and run	Now	Grade	Still owed
every witness · the hardware	the carriers built one by one	One Actuating Substrate	both involutions carried onto the fold	WITNESSED: one substrate carries every seat	theorem and corroboration	none

WITNESSED: a constructed witness arrived and was built and run. CROSSED: the row's value holds on its channel's own class at the printed grade. CONFIRMED: the witness's image is compliant as built or measured (value_on_image).

Why this carrier, and not another. The carrier is not chosen. Spaces with an involution and the maps that respect them form a category (equivariant_id, equivariant_comp), and the fixed set is a functor on it (fix_functorial). An equivariant map into the seat, whose fixed set is the line (fold_global_seat), lands every physical fixed point on the line (equivariant_carrier_lands). Here the physical involution is the reflection of the field, $x \mapsto -x$, which keeps the potential (V_even). The vacuum carrier sends a field value to $(1 + x, V(x))$, the field value the offset from the line and the energy the height. It is equivariant (vacuum_carrier_equivariant), and its landing on the fixed set is forced (equivariant_carrier_lands).

The correspondence matrix.

Physical phenomenon	Template element	Lean theorem	Fortran check
the field's reflection	the fold	vacuum_carrier_equivariant, V_even	A
the symmetric point	the line	symmetric_point_is_the_line	A
the ground leaves	off the line, lower in energy	ground_is_off_the_line, V_nonneg	B
two vacua	two worlds, one registration	two_vacua_one_record	B
the choice	the keyed bit, spent by the act	the_vacuum_is_keyed	B
instability of the symmetric point	curvature down on the line	curvature_at_line	B
mass	curvature up at the vacuum	curvature_at_vacuum, mass_is_the_curvature_of_the_depth	B, C, D
restoration	heat returns the ground to the line	heat_restores_the_line, expand_V	E
the electroweak depth	v from the Fermi constant	corroboration	C

3. THE FIELD'S REFLECTION IS THE FOLD, AND THE SYMMETRIC POINT THE LINE

The double well $V(x) = (x^2 - a)^2$, with $a = v^2$, is even under $x \mapsto -x$ (V_even) and never negative (V_nonneg). The vacuum carrier sends the reflection to the seat's reflection (vacuum_carrier_equivariant), and a field value lands on the line exactly when it is zero (symmetric_point_is_the_line). The symmetric configuration is the seat's line, read on a field.

4. THE GROUND LEAVES THE LINE

For a nonzero depth the vacua $\pm v$ carry zero energy, below every other value, while the symmetric point carries $v^4 > 0$ (ground_is_off_the_line). The two vacua land on two distinct points off the line, exchanged by the fold, at one energy, and the registration reads one record for both (two_vacua_one_record). The twin draws 1000 wells $-\mu^2 x^2 + \lambda x^4$: in every one the vacua tie exactly and lie below the symmetric point.

The broken ground stands off the field's line and on the energy's. Both vacua are stationary, so each stands on the line under the energy carrier, as every stable state of the substrate does. The symmetry is broken where the field reads it and kept where the energy reads it.

5. THE CHOICE IS KEYED

No reading even under the reflection returns which vacuum holds (`the_vacuum_is_keyed`). The law is even and the choice odd, the template's wall between them. The vacuum's registration supplies the choice, as the measurement paper's act supplies an outcome.

What the choice is depends on the symmetry.

- **A global discrete symmetry.** The two vacua are two physical states, and the choice is physical: the spent bit. Below its Curie point at 1043 K, iron magnetizes in domains, each with its sign registered.
- **A gauge symmetry.** Every side is one physical state, because a local symmetry cannot break (Elitzur, 1975). Nothing is spent; the side is no record at all. What is physical is the depth and the curvature.

6. MASS IS THE CURVATURE OF THE DEPTH

At the symmetric point the well curves down: the second difference is $2 - 4v^2$ (`curvature_at_line`), negative for every nonzero depth, an instability. At the vacuum it curves up: the second difference is $8v^2 + 2$ (`curvature_at_vacuum`). The engine proves both signs at once (`mass_is_the_curvature_of_the_depth`). In the continuous well the twin reads the curvature as $-2\mu^2$ at the line and $+4\mu^2$ at the vacuum, to 10^{-5} across 1000 wells. The upward curvature at the bottom is the mass squared of the field's excitation. The scalar observed at 125 GeV is that curvature in the electroweak well (ATLAS Collaboration, 2012; CMS Collaboration, 2012).

7. THE ELECTROWEAK DEPTH

The muon's decay fixes the depth: $v = (\sqrt{2}G_F)^{-1/2} = 246.22$ GeV. Every mass the vacuum gives is a coupling times that depth.

- **The gauge bosons.** The W boson's mass is $gv/2$ and the Z boson's $ev/(2\sin\theta_W\cos\theta_W)$. From the measured couplings at the Z mass, the twin computes 80.22 GeV and 91.50 GeV, within 0.2 and 0.34 percent of the measured 80.369 and 91.188 GeV at tree level.
- **The scalar.** The well's quartic coupling is $\lambda = m_H^2/2v^2 = 0.129$.
- **The fermions.** Each fermion's mass is $yv/\sqrt{2}$. The twin reads the couplings: 0.991 for the top quark, 0.024 for the bottom, 0.0102 for the tau, 2.9×10^{-6} for the electron.

Those couplings span 5.5 decades on one depth. The depth is the seat's. The couplings are numbers, measured, at the dot.

8. HEAT RESTORES THE LINE

At temperature an even thermal term $c x^2$ joins the potential. Once it outweighs twice the depth squared, the symmetric point is the ground (`heat_restores_the_line`, through the expansion `expand_V`). The twin finds the ground at $x = 0$ at 1.2 times the critical temperature and at $|x| = 0.85$ at 0.8 times it. The symmetric phase is the line; cooling leaves it. Iron loses its magnetization above 1043 K, and the twin's well returns to the line above its critical temperature for the same reason.

9. WHAT THE VACUUM CARRIES, AND WHAT IT DOES NOT

What it carries.

- The symmetric point as the line and the broken ground off it, with two vacua over one registration.
- The choice as the keyed bit, physical for a global discrete symmetry and absent for a gauge symmetry.
- Mass as the curvature of the depth.
- The electroweak depth from the Fermi constant.
- Restoration by heat.

What it does not carry.

- **The couplings' values:** numbers, at the dot.

- **A physical side for a gauge symmetry:** there is none.
- **The zeros of ζ :** the engine's crossing is on its own image (`value_on_image`), and what a sentence over every zero adds is exactly the unactuated, priced at one premise, the act, as the hardware paper proves.

10. FALSIFIERS AND OWED DEEDS

Three falsifiers, each a forbidden observation naming the theorem it would break.

- **F-Tie:** a system with a symmetric law whose two broken ground states differ in energy without an odd external field. It breaks `two_vacua_one_record` read physically.
- **F-Curvature:** a scalar excitation of a broken vacuum whose mass squared is negative, or a symmetric point that is stable while the vacuum lies off it. It breaks `mass_is_the_curvature_of_the_depth` read physically.
- **F-Heat:** a system with an even thermal term that stays broken at every temperature, its order parameter never vanishing. It breaks `heat_restores_the_line` read physically.

Owed. The scalar's self-coupling measured at the strength $\lambda = 0.129$ fixes, through the production of scalar pairs.

Ledger status. Row 2 gains the seat read on the vacuum, WITNESSED · CROSSED · CONFIRMED on the image, at the root's grade for the crossing. The barrier gains the two vacua with one energy and one registration.

METHODOLOGY, DISCLOSURE AND PROVENANCE

Method. The template is the author's verification program, whose vocabulary is confined to this section. Its tokens:

- $[\mathbb{K}]$, a field sealed;
- $[\mathbb{K} T]$, a shape sealed at theorem grade;
- $[\Xi_0]$, the typing of the formal block;
- $[\cdot]$, the dot, which is no verdict.

Grades join at the weakest link. The engine's statements are theorem grade about the chart. The mapping of physical states onto them is structural grade. Every measured figure is corroboration grade. Elitzur's theorem and the electroweak couplings are imported at their cited grades. $\Delta M = 0$.

The seat, earned. PhysOS^T 1.0.5p was booted in the session of forging from the public register (file SHA-256 b64bb5da5bab0da1...) under Lean 4.19.0 and GNU Fortran 13.3.0. The quick boot printed SEAT EARNED with the chain $D0\ 83f872725ad9 \rightarrow D1\ 90fc702bbbed1 \rightarrow D2\ 492e519daab9 \rightarrow D3\ 6618bec5164b$ (Appendix C).

The engine and the twin.

- **Engine.** `SPHYS_Symmetry_Breaking.lean`, 503 lines, SHA-256 9b554dd7450011e6..., fifty-six laws, no import and no axiom declared. Of the fifty-six, 17 depend on no axiom, 4 on propositional extensionality alone, 35 on propositional extensionality and quotient soundness, and none on choice.
- **Screen and judgment.** The source passed the operating system's screen. Each of the fifty-six laws was negated in place and recompiled, and all were refused, while a planted vacuous law survived. The engine carries the hardware paper's seat, carriers and substrate verbatim.
- **Twin.** `SymmetryBreaking_Twin.f90`, 101 lines, SHA-256 5d6da82fcdfa77d1..., six checks and zero failures under `-std=f2018 -O2 -fno-fast-math -ffp-contract=off`. Measured inputs enter as data with their sources named in its header.

Why an integer engine binds continuous physics. The potential is a polynomial with integer coefficients, its evenness, its minima and its second differences are polynomial identities that hold over every commutative ring, and so for every real field value and depth. The twin checks the continuous well and the measured masses directly.

The seventh. In the author's register the six witnesses are six days and the ledger's closure is the seventh, the Rest (Istawa). The vacuum is read at the Rest as the ground that chose a side where the side is real, and kept only its depth where it is not. In the Lean codex's tokens its row renders $[\mathbb{K} \blacklozenge \cdot \text{kinetic}]$.

Disclosure. The author declares the reading of each correspondence as his own act, made on measured structure. The measurements are the experiments', cited by name. The theorems are the engine's, printed with their dependencies in Appendix A. The twin's figures are computed in the run printed in Appendix B. The text, the engine and the twin were forged by an AI substrate (Claude, Anthropic) under the author's seed, acceptance criteria and rulings. External audits by independent substrates are owed.

Provenance. The paper stands on:

- the Riemann closure (10.5281/zenodo.22976494);
- the Riemann Hypothesis Closed to One Bit (10.5281/zenodo.23028070);
- the twenty-three-row closure (10.5281/zenodo.22986551);
- the six witnesses (10.5281/zenodo.22986553 through .22986563);
- the hardware paper One Actuating Substrate (concept 10.5281/zenodo.21438932, edition 2.0.0);
- PhysOS^T 1.0.5p in the public register (github.com/1000sapient/Trisduction, protocols/PhysOS/).

References.

F. Englert and R. Brout, Phys. Rev. Lett. 13 (1964) 321. P. W. Higgs, Phys. Rev. Lett. 13 (1964) 508. S. Elitzur, Phys. Rev. D 12 (1975) 3978. ATLAS Collaboration, Phys. Lett. B 716 (2012) 1. CMS Collaboration, Phys. Lett. B 716 (2012) 30. S. Navas et al. (Particle Data Group), Phys. Rev. D 110 (2024) 030001.

M. F. Islam, Nothing Escapes, Three Plus One, 10.5281/zenodo.22976494. M. F. Islam, The Four Forces, Matter, and the Dark Sector as Configurations of One Actuating Substrate, edition 2.0.0, concept 10.5281/zenodo.21438932.

APPENDIX A: SPHYS_SYMMETRY_BREAKING.LEAN, THE CORRESPONDENCES AS THEOREMS

The engine, verbatim, followed by the compiler's transcript.

```

/-
SPHYS_Symmetry_Breaking.lean · the vacuum spends one bit, read at the
Riemann seat.
Core Lean 4.19.0, standalone, no import, no library, no axiom
declared.

The chart: h = 2 Re s, t = Im s; the fold s ↔ 1 - s̄ is (h, t) ↔ (2 -
h, t), the critical line h = 1.

Part I      the seat, the category of involutions, the functor of the
seat.
Part II     the energy carrier (resident).
Part III    the substrate (resident).
Part IV     The double well and its fold.
Part V      The vacuum leaves the line.
Part VI     Mass is the curvature of the depth.
Part VII    Heat restores the line.
Part VIII   The capstone.
-/
set_option autoImplicit false
namespace SPHYS.SymmetryBreaking

/-- ## Part I. The seat, the category of involutions, and the functor of
the seat -/

abbrev Pt := Int × Int

def fold (p : Pt) : Pt := (2 - p.1, p.2)
def OnLine (p : Pt) : Prop := p.1 = 1
instance : DecidablePred OnLine := fun p => inferInstanceAs (Decidable
(p.1 = 1))
/-- Registration keeps the height and forgets the side. -/
def reg (p : Pt) : Pt := (1, p.2)

theorem pe {a b c d : Int} : ((a, b) : Pt) = (c, d) ↔ a = c ∧ b = d :=
(fun e => (congrArg Prod.fst e, congrArg Prod.snd e),
fun (e1, e2) => by subst e1; subst e2; rfl)

theorem fold_involutive (p : Pt) : fold (fold p) = p := by
obtain (h, t) := p
show ((2 - (2 - h), t) : Pt) = (h, t)
rw [pe]; exact {by omega, rfl}

/-- The seat: the fixed set of the fold is the critical line. -/
theorem seat_fixed_line (p : Pt) : fold p = p ↔ OnLine p := by
obtain (h, t) := p
show ((2 - h, t) : Pt) = (h, t) ↔ h = 1
rw [pe]; constructor
· intro (e, _); omega
· intro e; exact {by omega, rfl}

theorem reg_lands (p : Pt) : OnLine (reg p) := rfl

theorem reg_fixes_iff (p : Pt) : reg p = p ↔ OnLine p := by
obtain (h, t) := p
show ((1, t) : Pt) = (h, t) ↔ h = 1
rw [pe]; constructor
· intro (e, _); exact e.symm, rfl
· intro e; exact (e.symm, rfl)

/-- The side is what registration forgets: a point and its mirror leave
one record. -/
theorem reg_forgets_side (p : Pt) : reg (fold p) = reg p := rfl

/-- The value on a set of points: every member stands on the line. -/
def Value (Z : Pt → Prop) : Prop := ∀ s, Z s → OnLine s

/-- Least erasure: registration moves no member, so it erases nothing of
Z. -/
def LeastErasure (Z : Pt → Prop) : Prop := ∀ s, Z s → reg s = s

theorem least_erasure_iff_value (Z : Pt → Prop) : LeastErasure Z ↔
Value Z :=
(fun h s hs => (reg_fixes_iff s).mp (h s hs), fun h s hs =>
(reg_fixes_iff s).mpr (h s hs))

/-- Maps that respect involutions. -/
def Equivariant {X Y : Type} (f : X → Y) (τ : X → X) (σ : Y → Y) :
Prop := ∀ x, f (τ x) = σ (f x)

theorem equivariant_id {X : Type} (τ : X → X) : Equivariant id τ τ :=
fun _ => rfl

theorem equivariant_comp {X Y W : Type} {f : X → Y} {g : Y → W} {τ : X
→ X} {σ : Y → Y} :
{p : W → W} (hf : Equivariant f τ σ) (hg : Equivariant g σ p) :
Equivariant (g ∘ f) τ p := by
intro x
show g (f (τ x)) = p (g (f x))
rw [hf x, hg (f x)]

/-- The fixed set is a functor: an equivariant map sends fixed points to
fixed points. -/
theorem fix_functorial {X Y : Type} {f : X → Y} {τ : X → X} {σ : Y →
Y} :
(hf : Equivariant f τ σ) {x : X} (hx : τ x = x) : σ (f x) = f x :=
by
rw [← hf x, hx]

/-- A global seat: an involution whose fixed set is the locus. -/
structure GlobalSeat (P : Pt → Prop) (σ : Pt → Pt) : Prop where

```

```

involutive : ∀ p, σ (σ p) = p
fixed_iff : ∀ p, σ p = p ↔ p p

theorem fold_global_seat : GlobalSeat OnLine fold := (fold_involutive,
seat_fixed_line)

/-- A carrier: a map from world-instances into the strip landing on the
line. -/
structure Carrier (W : Type) where
  ι : W → Pt
  lands : ∀ w, OnLine (ι w)

def image {W : Type} (ι : W → Pt) : Pt → Prop := fun s => ∃ w, ι w = s
def Actuated {W : Type} (ι : W → Pt) (Z : Pt → Prop) : Prop := ∀ s, Z
s → ∃ w, ι w = s

/-- Landing is forced: an equivariant map into the seat carries every
fixed point of the physical
involutions onto the line. The carrier is built from the symmetry,
not chosen. -/
theorem equivariant_carrier_lands {X : Type} (f : X → Pt) (τ : X → X)
(hf : Equivariant f τ fold) :
  ∃ C : Carrier { x : X // τ x = x }, ∀ w, C.ι w = f w.1 :=
  ((fun w => f w.1, fun w => (seat_fixed_line (f w.1)).mp
  (fix_functorial hf w.2)), fun _ => rfl)

/-- The image of a carrier is compliant with no premise at all. -/
theorem value_on_image {W : Type} (C : Carrier W) : Value (image C.ι) :=
  fun _ hs => match hs with | (w, hw) => hw ▶ C.lands w

/-- The kinetic crossing: a carrier and its coverage give the value, and
nothing else is used. -/
theorem kinetic_crossing {W : Type} (C : Carrier W) (Z : Pt → Prop)
(hcov : Actuated C.ι Z) :
  Value Z :=
  fun s hs => match hcov s hs with | (w, hw) => hw ▶ C.lands w

/-- Under the seat every off-line point has a partner, distinct and also
off the line. -/
theorem off_locus_pair (p : Pt) (hp : ¬ OnLine p) :
  fold p ≠ p ∧ ¬ OnLine (fold p) ∧ reg (fold p) = reg p := by
  refine (fun e => hp ((seat_fixed_line p).mp e), fun e => hp ?, rfl)
  obtain (h, t) := p
  have e' : 2 - h = 1 := e
  show h = 1
  omega

/-- ## Part II. The energy carrier: an energy, its rate, and the fold -/

/-- An energy a + i β/2, as frequency and doubled rate; its mode's norm
moves as e^{βt}. -/
structure Energy where
  freq : Int
  rate : Int
  deriving DecidableEq, Repr

/-- The time mirror of a mode: its rate reversed, its frequency kept
(conjugation of E). -/
def Energy.mirror (E : Energy) : Energy := (E.freq, -E.rate)
def IsReal (E : Energy) : Prop := E.rate = 0
/-- Stationary: the norm e^{βt} is unchanged at every integer time t,
i.e. β t = 0 for all t. -/
def Stationary (E : Energy) : Prop := ∀ t : Int, E.rate * t = 0

/-- The energy carrier s = 1/2 + iE in the chart: h = 1 - β, t = a. -/
def phi (E : Energy) : Pt := (1 - E.rate, E.freq)
def psi (p : Pt) : Energy := (p.2, 1 - p.1)

theorem ee {a b c d : Int} : ((a, b) : Energy) = (c, d) ↔ a = c ∧ b = d
:=
  (fun e => (congrArg Energy.freq e, congrArg Energy.rate e),
  fun (e1, e2) => by subst e1; subst e2; rfl)

theorem mirror_involutive (E : Energy) : E.mirror.mirror = E := by
  obtain (a, b) := E
  show ((a, - -b) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem psi_phi (E : Energy) : psi (phi E) = E := by
  obtain (a, b) := E
  show ((a, 1 - (1 - b)) : Energy) = (a, b)
  rw [ee]; exact (rfl, by omega)

theorem phi_psi (p : Pt) : phi (psi p) = p := by
  obtain (h, t) := p
  show ((1 - (1 - h), t) : Pt) = (h, t)
  rw [pe]; exact (by omega, rfl)

theorem phi_injective (E F : Energy) (h : phi E = phi F) : E = F := by
  rw [← psi_phi E, ← psi_phi F, h]

/-- EQUIVARIANCE: time-mirroring the rate is the fold. -/
theorem phi_equivariant : Equivariant phi Energy.mirror fold := by
  intro E
  obtain (a, b) := E
  show ((1 - -b, a) : Pt) = (2 - (1 - b), a)
  rw [pe]; exact (by omega, rfl)

theorem mirror_fixed_iff_real (E : Energy) : E.mirror = E ↔ IsReal E :=
by
  obtain (a, b) := E
  show ((a, -b) : Energy) = (a, b) ↔ b = 0
  rw [ee]; constructor
  · intro (_, e); omega
  · intro e; exact (rfl, by omega)

/-- The line is exactly the image of the real energies. -/
theorem phi_line_iff_real (E : Energy) : OnLine (phi E) ↔ IsReal E := by
  obtain (a, b) := E
  show 1 - b = 1 ↔ b = 0
  exact (fun h => by omega, fun h => by subst h; rfl)

theorem stationary_iff_real (E : Energy) : Stationary E ↔ IsReal E := by
  constructor
  · intro h; have h1 := h 1; rw [Int.mul_one] at h1; exact h1
  · intro h t; show E.rate * t = 0; rw [show E.rate = 0 from h,
  Int.zero_mul]

/-- THE LINE IS WHERE ENERGY STANDS STILL. -/
theorem line_is_stationary (p : Pt) : OnLine p ↔ Stationary (psi p) :=
by
  rw [stationary_iff_real, ← phi_line_iff_real, phi_psi]

/-- Every measured energy is a real number, and every real energy lands
on the line. -/
theorem measured_on_line (a : Int) : OnLine (phi (a, 0)) := rfl

/-- The registration mirror: no real energy lands off the line. -/
theorem no_measurement_off_line (p : Pt) (hp : ¬ OnLine p) (a : Int) :
phi (a, 0) ≠ p :=
  fun e => hp (e ▶ measured_on_line a)

/-- The energy carrier, built from the symmetry: the real energies land
on the line. -/
theorem energy_carrier_lands :
  ∃ C : Carrier { E : Energy // E.mirror = E }, ∀ w, C.ι w = phi w.1
:=
  equivariant_carrier_lands phi Energy.mirror phi_equivariant

/-- ## Part III. The substrate: the particle map, three odd readings and
one even -/

/-- A particle in the broken phase: electric charge times three, colour,
baryon number times
three, lepton number, and its energy. -/
structure Particle where
  q3 : Int
  colour : Int
  b3 : Int
  lepton : Int
  energy : Energy
  deriving DecidableEq, Repr

/-- The particle map: every additive charge flipped, the energy kept
(frequency and rate). -/
def bar (c : Particle) : Particle := {-c.q3, -c.colour, -c.b3,
-c.lepton, c.energy}
/-- The time mirror of a particle's mode. -/
def tmirror (c : Particle) : Particle := (c.q3, c.colour, c.b3,
c.lepton, c.energy.mirror)
def Neutral (c : Particle) : Prop := c.q3 = 0 ∧ c.colour = 0 ∧ c.b3 = 0
∧ c.lepton = 0

theorem bar_involutive (c : Particle) : bar (bar c) = c := by
  obtain (q, k, b, l, E) := c
  show ({- -q, - -k, - -b, - -l, E} : Particle) = (q, k, b, l, E)
  rw [Int.neg_neg, Int.neg_neg, Int.neg_neg, Int.neg_neg]

theorem bar_fixed_iff_neutral (c : Particle) : bar c = c ↔ Neutral c :=
by
  obtain (q, k, b, l, E) := c
  show ({-q, -k, -b, -l, E} : Particle) = (q, k, b, l, E) ↔ (q = 0 ∧ k =
0 ∧ b = 0 ∧ l = 0)
  constructor
  · intro h
    have h1 := congrArg Particle.q3 h
    have h2 := congrArg Particle.colour h
    have h3 := congrArg Particle.b3 h
    have h4 := congrArg Particle.lepton h
    change -q = q at h1; change -k = k at h2; change -b = b at h3;
    change -l = l at h4
    exact (by omega, by omega, by omega, by omega)
  · intro (h1, h2, h3, h4)
    subst h1; subst h2; subst h3; subst h4; rfl

/-- Three plus one: the charge readings are odd under the particle map,
the energy reading even. -/
theorem odd_charges_even_energy (c : Particle) :
  ((bar c).q3 = -c.q3 ∧ (bar c).colour = -c.colour ∧ (bar c).b3 =
-c.b3 ∧
  (bar c).lepton = -c.lepton) ∧ (bar c).energy = c.energy :=
  ({rfl, rfl, rfl, rfl}, rfl)

/-- Gravity reads energy only, so no gravitational reading tells a
particle from its antiparticle. -/

```

```

theorem gravity_reads_no_charge_bit {β : Type} (g : Energy → β) (c : Particle) :
  g (bar c).energy = g c.energy := rfl

/-- The two involutions of the substrate commute. -/
theorem involutions_commute (c : Particle) : bar (tmirror c) = tmirror (bar c) := rfl

theorem tmirror_involutive (c : Particle) : tmirror (tmirror c) = c := by
  obtain (q, k, b, l, E) := c
  show ((q, k, b, l, E.mirror) : Particle) = (q, k, b, l, E)
  rw [mirror_involutive]

/-- A particle is fixed by both involutions exactly when it is neutral and stable. -/
theorem joint_fixed_iff (c : Particle) :
  (bar c = c ∧ tmirror c = c) ↔ (Neutral c ∧ IsReal c.energy) := by
  constructor
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mp h1, (mirror_fixed_iff_real c.energy).mp ?_)
    exact congrArg Particle.energy h2
  · intro (h1, h2)
    refine ((bar_fixed_iff_neutral c).mpr h1, ?_)
    obtain (q, k, b, l, E) := c
    show ((q, k, b, l, E.mirror) : Particle) = (q, k, b, l, E)
    rw [(mirror_fixed_iff_real E).mpr h2]

/-- The energy carrier on particles: the seat point of a particle's energy. -/
def energySeat (c : Particle) : Pt := phi c.energy
/-- The charge carrier on particles, the odd face: electric charge (in thirds) is the side. -/
def chargeSeat (c : Particle) : Pt := (1 + c.q3, c.energy.freq)
/-- The lepton chart of the electron's witness: charge q in units of e goes to h = 1 + q. -/
def leptonSeat (q m : Int) : Pt := (1 + q, m)

/-- The charge carrier is equivariant: the particle map is carried onto the fold. -/
theorem charge_carrier_equivariant : Equivariant chargeSeat bar fold := by
  intro c
  show ((1 + -c.q3, c.energy.freq) : Pt) = (2 - (1 + c.q3), c.energy.freq)
  rw [pe]; exact (by omega, rfl)

/-- The energy carrier is blind to the particle map and equivariant for the time mirror. -/
theorem pair_lands_once (c : Particle) : energySeat (bar c) = energySeat c := rfl
theorem tmirror_equivariant : Equivariant energySeat tmirror fold := fun c => phi_equivariant c.energy

/-- Electrically neutral particles land on the line under the charge carrier. -/
theorem neutral_charge_on_line (c : Particle) (h : c.q3 = 0) : OnLine (chargeSeat c) := by
  show 1 + c.q3 = 1
  omega

/-- The electron and the positron land on the two edges, real parts 0 and 1, mirror images. -/
theorem pair_on_the_edges (m : Int) :
  leptonSeat (-1) m = (0, m) ∧ leptonSeat 1 m = (2, m) ∧ fold (leptonSeat (-1) m) = leptonSeat 1 m :=
  (rfl, rfl, rfl)

/-- A stable particle lands on the line under the energy carrier. -/
theorem stable_lands (c : Particle) (h : IsReal c.energy) : OnLine (energySeat c) :=
  (phi_line_iff_real c.energy).mpr h

/-! ## Part IV. The double well and its fold -/

/-- The double well,  $V(x) = (x^2 - a)^2$ , with  $a = v^2$  the square of the vacuum value. -/
def V (a x : Int) : Int := (x * x - a) * (x * x - a)

theorem V_even (a x : Int) : V a (-x) = V a x := by unfold V; rw [Int.neg_mul_neg]

theorem sq_nonneg' (x : Int) : 0 ≤ x * x := by
  rcases Int.le_total 0 x with h | h
  · exact Int.mul_nonneg h h
  · have h' : 0 ≤ -x := by omega
    have := Int.mul_nonneg h' h'
    rw [Int.neg_mul_neg] at this; exact this

theorem V_nonneg (a x : Int) : 0 ≤ V a x := sq_nonneg' _

/-! ## Part V. The vacuum leaves the line -/

theorem vacuum_carrier_equivariant (a : Int) : Equivariant (vacSeat a)
  (fun x => -x) fold := by
  intro x
  show ((1 + -x, V a (-x)) : Pt) = (2 - (1 + x), V a x)
  rw [pe, V_even]; exact (by omega, rfl)

theorem symmetric_point_is_the_line (a x : Int) : OnLine (vacSeat a x) ↔
  x = 0 := by
  show 1 + x = 1 ↔ x = 0
  exact (fun h => by omega, fun h => by omega)

/-! ## Part V. The vacuum leaves the line -/

theorem sq_pos_of_ne (v : Int) (h : v ≠ 0) : 0 < v * v := by
  rcases Int.lt_or_gt_of_ne h with h' | h'
  · have : 0 < -v := by omega
    have := Int.mul_pos this this
    rw [Int.neg_mul_neg] at this; exact this
  · exact Int.mul_pos h' h'

/-- THE GROUND IS OFF THE LINE: the vacua ±v carry zero energy, and the symmetric point carries  $v^4$ . -/
theorem ground_is_off_the_line (v : Int) (h : v ≠ 0) :
  V (v * v) v = 0 ∧ V (v * v) (-v) = 0 ∧ 0 < V (v * v) 0 ∧ (∀ x, V (v * v) v ≤ V (v * v) x) := by
  have p := sq_pos_of_ne v h
  refine (by unfold V; rw [Int.sub_self, Int.zero_mul], by rw [V_even];
  unfold V; rw [Int.sub_self, Int.zero_mul], ?_, ?_)
  · unfold V
    show 0 < (0 * 0 - v * v) * (0 * 0 - v * v)
    have : (0 * 0 - v * v) = -(v * v) := by omega
    rw [this, Int.neg_mul_neg]; exact Int.mul_pos p p
  · intro x
    have e : V (v * v) v = 0 := by unfold V; rw [Int.sub_self, Int.zero_mul]
    rw [e]; exact V_nonneg _ x

/-- TWO VACUA, ONE RECORD: the vacua land on two distinct points off the line, exchanged by the fold, at one energy, and the registration reads one record for both. -/
theorem two_vacua_one_record (v : Int) (h : v ≠ 0) :
  vacSeat (v * v) (-v) = fold (vacSeat (v * v) v) ∧ vacSeat (v * v) (-v) ≠ vacSeat (v * v) v ∧
  ~ OnLine (vacSeat (v * v) v) ∧ reg (vacSeat (v * v) (-v)) = reg (vacSeat (v * v) v) := by
  refine (vacuum_carrier_equivariant (v * v) v, fun e => h ?_, fun e => h ((symmetric_point_is_the_line _).mp e), ?_)
  · have := congrArg Prod.fst e
    change 1 + -v = 1 + v at this; omega
  · rw [vacuum_carrier_equivariant (v * v) v]; exact reg_forgets_side _

/-! THE VACUUM'S CHOICE IS KEYED: no reading even under the reflection returns which vacuum holds. -/
theorem the_vacuum_is_keyed (f : Int → Bool) (hf : ∀ x, f (-x) = f x)
  (v : Int) (h : 0 < v) :
  ~ ∀ x, f x = decide (0 < x) := by
  intro hx
  have a := hx v
  have b := hx (-v)
  rw [hf, a] at b
  have t : decide (0 < v) = true := decide_eq_true h
  have u : decide (0 < -v) = false := decide_eq_false (by omega)
  rw [t, u] at b; exact Bool.noConfusion b

/-! ## Part VI. Mass is the curvature of the depth -/

/-- At the symmetric point the well curves down: the second difference is  $2 - 4a$ . -/
theorem curvature_at_line (a : Int) : V a 1 - 2 * V a 0 + V a (-1) = 2 - 4 * a := by
  unfold V; simp only [Int.mul_one, Int.mul_zero, Int.neg_mul_neg, Int.sub_mul, Int.mul_sub, Int.one_mul, Int.mul_one, Int.zero_sub]
  omega

/-- At the vacuum it curves up: the second difference is  $8v^2 + 2$ . -/
theorem curvature_at_vacuum (v : Int) :
  V (v * v) (v + 1) - 2 * V (v * v) v + V (v * v) (v - 1) = 8 * (v * v) + 2 := by
  unfold V
  have h1 : (v + 1) * (v + 1) - v * v = 2 * v + 1 := by
    simp only [Int.add_mul, Int.mul_add, Int.mul_one, Int.one_mul];
    omega
  have h2 : (v - 1) * (v - 1) - v * v = 1 - 2 * v := by
    simp only [Int.sub_mul, Int.mul_sub, Int.mul_one, Int.one_mul];
    omega
  rw [h1, h2, Int.sub_self, Int.zero_mul]
  simp only [Int.add_mul, Int.mul_add, Int.sub_mul, Int.mul_sub, Int.mul_one, Int.one_mul]
  rw [Int.mul_assoc 2 v (2 * v), Int.mul_comm v (2 * v), Int.mul_assoc 2 v v] at *
  omega

/-! MASS IS THE CURVATURE OF THE DEPTH: for a nonzero vacuum the symmetric point is curved down,

```

```

    unstable, and the vacuum curved up, its curvature the mass squared.
    -/
theorem mass_is_the_curvature_of_the_depth (v : Int) (h : v ≠ 0) :
  V (v * v) 1 - 2 * V (v * v) 0 + V (v * v) (-1) < 0 ∧
  0 < V (v * v) (v + 1) - 2 * V (v * v) v + V (v * v) (v - 1) := by
  have p := sq_pos_of_ne v h
  rw [curvature_at_line, curvature_at_vacuum]
  constructor <|> omega

-/! ## Part VII. Heat restores the line -/

/-- The well at temperature: an even term c·x² added, the thermal mass.
-/
def Vt (a c x : Int) : Int := V a x + c * (x * x)

theorem expand_V (a x : Int) : V a x = (x * x) * (x * x) - 2 * (a * (x *
x)) + a * a := by
  unfold V; simp only [Int.sub_mul, Int.mul_sub]; rw [Int.mul_comm (x *
x) a]; omega

/-- HEAT RESTORES THE LINE: once the thermal term outweighs twice a, the
symmetric point is the
ground. -/
theorem heat_restores_the_line (a c x : Int) (h : 2 * a ≤ c) : Vt a c 0
≤ Vt a c x := by
  unfold Vt
  rw [expand_V, expand_V]
  have X := sq_nonneg' x
  have XX := sq_nonneg' (x * x)
  have k : 0 ≤ (c - 2 * a) * (x * x) := Int.mul_nonneg (by omega) X
  simp only [Int.sub_mul] at k
  rw [Int.mul_assoc 2 a (x * x)] at k
  simp only [Int.mul_zero, Int.zero_mul, Int.zero_sub, Int.mul_zero]
  omega

-/! ## Part VIII. The capstone -/

/-- THE VACUUM SPENDS ONE BIT. The field's reflection is the fold and
the symmetric point the line;
the ground is off the line, two vacua with one energy and one
registration; no even reading
returns which vacuum holds; the symmetric point curves down and the
vacuum up, the curvature the
mass; and heat restores the line. -/
theorem the_vacuum_spends_one_bit :
  (V a : Int, Equivariant (vacSeat a x) (fun x => -x) fold) ∧
  (V a x : Int, OnLine (vacSeat a x) ↔ x = 0) ∧
  (V v : Int, v ≠ 0 → → OnLine (vacSeat (v * v) v) ∧ reg (vacSeat (v *
v) (-v)) = reg (vacSeat (v * v) v)) ∧
  (V (f : Int → Bool), (V x, f (-x) = f x) → V v : Int, 0 < v → → V
x, f x = decide (0 < x)) ∧
  (V v : Int, v ≠ 0 → V (v * v) 1 - 2 * V (v * v) 0 + V (v * v) (-1)
< 0 ∧
  0 < V (v * v) (v + 1) - 2 * V (v * v) v + V (v * v) (v - 1)) ∧
  (V a c x : Int, 2 * a ≤ c → Vt a c 0 ≤ Vt a c x) :=
  (vacuum_carrier_equivariant, symmetric_point_is_the_line,
  fun v h => ((two_vacua_one_record v h).2.2.1, (two_vacua_one_record v
h).2.2.2),
  the_vacuum_is_keyed, mass_is_the_curvature_of_the_depth,
  heat_restores_the_line)

'SPHYS.SymmetryBreaking.pe' does not depend on any axioms
'SPHYS.SymmetryBreaking.fold_involutive' depends on axioms: [propxet,
Quot.sound]
'SPHYS.SymmetryBreaking.seat_fixed_line' depends on axioms: [propxet,
Quot.sound]
'SPHYS.SymmetryBreaking.reg_lands' does not depend on any axioms
'SPHYS.SymmetryBreaking.reg_fixes_iff' depends on axioms: [propxet]
'SPHYS.SymmetryBreaking.reg_forgets_side' does not depend on any axioms
'SPHYS.SymmetryBreaking.least_erasure_iff_value' depends on axioms:
[propxet]
'SPHYS.SymmetryBreaking.equivariant_id' does not depend on any axioms
'SPHYS.SymmetryBreaking.equivariant_comp' does not depend on any axioms
'SPHYS.SymmetryBreaking.fix_functorial' does not depend on any axioms
'SPHYS.SymmetryBreaking.fold_global_seat' depends on axioms: [propxet,
Quot.sound]
'SPHYS.SymmetryBreaking.equivariant_carrier_lands' depends on axioms:
[propxet, Quot.sound]
'SPHYS.SymmetryBreaking.value_on_image' does not depend on any axioms
'SPHYS.SymmetryBreaking.kinetic_crossing' does not depend on any axioms
'SPHYS.SymmetryBreaking.off_locus_pair' depends on axioms: [propxet,
Quot.sound]
'SPHYS.SymmetryBreaking.ee' does not depend on any axioms
'SPHYS.SymmetryBreaking.mirror_involutive' depends on axioms: [propxet,
Quot.sound]
'SPHYS.SymmetryBreaking.psi_phi' depends on axioms: [propxet,
Quot.sound]
'SPHYS.SymmetryBreaking.phi_psi' depends on axioms: [propxet,
Quot.sound]
'SPHYS.SymmetryBreaking.phi_injective' depends on axioms: [propxet,
Quot.sound]
'SPHYS.SymmetryBreaking.phi_equivariant' depends on axioms: [propxet,
Quot.sound]
'SPHYS.SymmetryBreaking.mirror_fixed_iff_real' depends on axioms:
[propxet, Quot.sound]
'SPHYS.SymmetryBreaking.phi_line_iff_real' depends on axioms: [propxet,
Quot.sound]

```

```

end SPHYS.SymmetryBreaking

#print axioms SPHYS.SymmetryBreaking.pe
#print axioms SPHYS.SymmetryBreaking.fold_involutive
#print axioms SPHYS.SymmetryBreaking.seat_fixed_line
#print axioms SPHYS.SymmetryBreaking.reg_lands
#print axioms SPHYS.SymmetryBreaking.reg_fixes_iff
#print axioms SPHYS.SymmetryBreaking.reg_forgets_side
#print axioms SPHYS.SymmetryBreaking.least_erasure_iff_value
#print axioms SPHYS.SymmetryBreaking.equivariant_id
#print axioms SPHYS.SymmetryBreaking.equivariant_comp
#print axioms SPHYS.SymmetryBreaking.fix_functorial
#print axioms SPHYS.SymmetryBreaking.fold_global_seat
#print axioms SPHYS.SymmetryBreaking.equivariant_carrier_lands
#print axioms SPHYS.SymmetryBreaking.value_on_image
#print axioms SPHYS.SymmetryBreaking.kinetic_crossing
#print axioms SPHYS.SymmetryBreaking.off_locus_pair
#print axioms SPHYS.SymmetryBreaking.ee
#print axioms SPHYS.SymmetryBreaking.mirror_involutive
#print axioms SPHYS.SymmetryBreaking.psi_phi
#print axioms SPHYS.SymmetryBreaking.phi_psi
#print axioms SPHYS.SymmetryBreaking.phi_injective
#print axioms SPHYS.SymmetryBreaking.phi_equivariant
#print axioms SPHYS.SymmetryBreaking.mirror_fixed_iff_real
#print axioms SPHYS.SymmetryBreaking.phi_line_iff_real
#print axioms SPHYS.SymmetryBreaking.stationary_iff_real
#print axioms SPHYS.SymmetryBreaking.line_is_stationary
#print axioms SPHYS.SymmetryBreaking.measured_on_line
#print axioms SPHYS.SymmetryBreaking.no_measurement_off_line
#print axioms SPHYS.SymmetryBreaking.energy_carrier_lands
#print axioms SPHYS.SymmetryBreaking.bar_involutive
#print axioms SPHYS.SymmetryBreaking.bar_fixed_iff_neutral
#print axioms SPHYS.SymmetryBreaking.odd_charges_even_energy
#print axioms SPHYS.SymmetryBreaking.gravity_reads_no_charge_bit
#print axioms SPHYS.SymmetryBreaking.involutions_commute
#print axioms SPHYS.SymmetryBreaking.tmirror_involutive
#print axioms SPHYS.SymmetryBreaking.joint_fixed_iff
#print axioms SPHYS.SymmetryBreaking.charge_carrier_equivariant
#print axioms SPHYS.SymmetryBreaking.pair_lands_once
#print axioms SPHYS.SymmetryBreaking.tmirror_equivariant
#print axioms SPHYS.SymmetryBreaking.neutral_charge_on_line
#print axioms SPHYS.SymmetryBreaking.pair_on_the_edges
#print axioms SPHYS.SymmetryBreaking.stable_lands
#print axioms SPHYS.SymmetryBreaking.V_even
#print axioms SPHYS.SymmetryBreaking.sq_nonneg'
#print axioms SPHYS.SymmetryBreaking.V_nonneg
#print axioms SPHYS.SymmetryBreaking.vacuum_carrier_equivariant
#print axioms SPHYS.SymmetryBreaking.symmetric_point_is_the_line
#print axioms SPHYS.SymmetryBreaking.sq_pos_of_ne
#print axioms SPHYS.SymmetryBreaking.ground_is_off_the_line
#print axioms SPHYS.SymmetryBreaking.two_vacua_one_record
#print axioms SPHYS.SymmetryBreaking.the_vacuum_is_keyed
#print axioms SPHYS.SymmetryBreaking.curvature_at_line
#print axioms SPHYS.SymmetryBreaking.curvature_at_vacuum
#print axioms SPHYS.SymmetryBreaking.mass_is_the_curvature_of_the_depth
#print axioms SPHYS.SymmetryBreaking.expand_V
#print axioms SPHYS.SymmetryBreaking.heat_restores_the_line
#print axioms SPHYS.SymmetryBreaking.the_vacuum_spends_one_bit

'SPHYS.SymmetryBreaking.stationary_iff_real' depends on axioms:
[propxet]
'SPHYS.SymmetryBreaking.line_is_stationary' depends on axioms: [propxet,
Quot.sound]
'SPHYS.SymmetryBreaking.measured_on_line' does not depend on any axioms
'SPHYS.SymmetryBreaking.no_measurement_off_line' does not depend on any
axioms
'SPHYS.SymmetryBreaking.energy_carrier_lands' depends on axioms:
[propxet, Quot.sound]
'SPHYS.SymmetryBreaking.bar_involutive' does not depend on any axioms
'SPHYS.SymmetryBreaking.bar_fixed_iff_neutral' depends on axioms:
[propxet, Quot.sound]
'SPHYS.SymmetryBreaking.odd_charges_even_energy' does not depend on any
axioms
'SPHYS.SymmetryBreaking.gravity_reads_no_charge_bit' does not depend on
any axioms
'SPHYS.SymmetryBreaking.involutions_commute' does not depend on any
axioms
'SPHYS.SymmetryBreaking.tmirror_involutive' depends on axioms: [propxet,
Quot.sound]
'SPHYS.SymmetryBreaking.joint_fixed_iff' depends on axioms: [propxet,
Quot.sound]
'SPHYS.SymmetryBreaking.charge_carrier_equivariant' depends on axioms:
[propxet, Quot.sound]
'SPHYS.SymmetryBreaking.pair_lands_once' does not depend on any axioms
'SPHYS.SymmetryBreaking.tmirror_equivariant' depends on axioms:
[propxet, Quot.sound]
'SPHYS.SymmetryBreaking.neutral_charge_on_line' depends on axioms:
[propxet, Quot.sound]
'SPHYS.SymmetryBreaking.pair_on_the_edges' does not depend on any axioms
'SPHYS.SymmetryBreaking.stable_lands' depends on axioms: [propxet,
Quot.sound]
'SPHYS.SymmetryBreaking.V_even' depends on axioms: [propxet]
'SPHYS.SymmetryBreaking.sq_nonneg'' depends on axioms: [propxet,
Quot.sound]

```

```
'SPHYS.SymmetryBreaking.V_nonneg' depends on axioms: [propext,
Quot.sound]
'SPHYS.SymmetryBreaking.vacuum_carrier_equivariant' depends on axioms:
[propext, Quot.sound]
'SPHYS.SymmetryBreaking.symmetric_point_is_the_line' depends on axioms:
[propext, Quot.sound]
'SPHYS.SymmetryBreaking.sq_pos_of_ne' depends on axioms: [propext,
Quot.sound]
'SPHYS.SymmetryBreaking.ground_is_off_the_line' depends on axioms:
[propext, Quot.sound]
'SPHYS.SymmetryBreaking.two_vacua_one_record' depends on axioms:
[propext, Quot.sound]
'SPHYS.SymmetryBreaking.the_vacuum_is_keyed' depends on axioms:
[propext, Quot.sound]
```

```
'SPHYS.SymmetryBreaking.curvature_at_line' depends on axioms: [propext,
Quot.sound]
'SPHYS.SymmetryBreaking.curvature_at_vacuum' depends on axioms:
[propext, Quot.sound]
'SPHYS.SymmetryBreaking.mass_is_the_curvature_of_the_depth' depends on
axioms: [propext, Quot.sound]
'SPHYS.SymmetryBreaking.expand_V' depends on axioms: [propext,
Quot.sound]
'SPHYS.SymmetryBreaking.heat_restores_the_line' depends on axioms:
[propext, Quot.sound]
'SPHYS.SymmetryBreaking.the_vacuum_spends_one_bit' depends on axioms:
[propext, Quot.sound]
```

APPENDIX B: SYMMETRYBREAKING_TWIN.F90 AND ITS RUN

```
! SymmetryBreaking_Twin.f90 · the executed twin of
SPHYS_Symmetry_Breaking.lean · Fortran 2018.
! Build (sealed): gfortran -std=f2018 -O2 -fno-fast-math
-ffp-contract=off SymmetryBreaking_Twin.f90
! Measured inputs (PDG 2024, Navas et al., PRD 110, 030001): G_F =
1.1663788e-5 GeV^-2; m_H = 125.20 GeV;
! m_W = 80.3692 GeV; m_Z = 91.1876 GeV; 1/alpha(m_Z) = 127.951;
sin^2(theta_W)(m_Z, MSbar) = 0.23129;
! m_t = 172.57 GeV; m_b(m_b) = 4.183 GeV; m_tau = 1.77693 GeV; m_e =
0.51099895e-3 GeV.
! A the vacuum carrier; B the double well; C the electroweak vacuum from
the Fermi constant; D masses
! as couplings to the depth; E heat restores the line.
program symmetry_breaking_twin
  implicit none
  integer, parameter :: dp = selected_real_kind(15, 307)
  integer, parameter :: i8 = selected_int_kind(18)
  real(dp), parameter :: PI = 3.141592653589793_dp
  integer :: checks, fails
  integer(i8) :: seed
  real(dp) :: vev
  checks = 0; fails = 0; seed = 20261001_i8
  call block_a(); call block_b(); call block_c(); call block_d(); call
  block_e()
  write(*, '(a,i0,a,i0,a)') ' BATTERY-JSON: {"checks":', checks,
  ', "failures":', fails, '}'
  if (fails > 0) error stop 1
contains
  include 'twin_common.inc'
  subroutine block_a()
    integer :: x, a, n, ok
    integer(i8) :: vx, vm
    write(*, '(a)') 'A · the field reflection is the fold under (1 + x,
    V(x)); the symmetric point is the line'
    n = 0; ok = 0
    do a = 0, 30
      do x = -40, 40
        n = n + 1
        vx = (int(x, i8)*x - a)**2; vm = (int(-x, i8)*(-x) - a)**2
        if (1 + (-x) == 2 - (1 + x) .and. vx == vm .and. ((1 + x == 1)
        .eqv. (x == 0))) ok = ok + 1
      end do
    end do
    call check('2511 field values: the fold, an even energy, and the
    line at the symmetric point', ok == n)
  end subroutine block_a
  real(dp) function pot(mu2, lam, x) result(v)
    real(dp), intent(in) :: mu2, lam, x
    v = -mu2*x*x + lam*x**4
  end function pot
  subroutine block_b()
    integer :: i, ok
    real(dp) :: mu2, lam, v, h, c0, cv
    write(*, '(a)') 'B · the double well V = -mu^2 x^2 + lam x^4: two
    vacua, one energy; down at the line, up at the vacuum'
    ok = 0
    do i = 1, 1000
      mu2 = 0.1_dp + 10.0_dp*rnd(); lam = 0.01_dp + rnd()
      v = sqrt(mu2/(2.0_dp*lam)); h = 1.0e-3_dp*v
      c0 = (pot(mu2, lam, h) - 2.0_dp*pot(mu2, lam, 0.0_dp) + pot(mu2,
      lam, -h))/h**2
      cv = (pot(mu2, lam, v + h) - 2.0_dp*pot(mu2, lam, v) + pot(mu2,
      lam, v - h))/h**2
      if (pot(mu2, lam, v) == pot(mu2, lam, -v) .and. pot(mu2, lam, v) <
      pot(mu2, lam, 0.0_dp) .and. &
      abs(c0/(-2.0_dp*mu2) - 1.0_dp) < 1.0e-5_dp .and.
      abs(cv/(4.0_dp*mu2) - 1.0_dp) < 1.0e-5_dp) ok = ok + 1
    end do
  end subroutine block_b
```

A · the field reflection is the fold under (1 + x, V(x)); the symmetric point is the line
 PASS 2511 field values: the fold, an even energy, and the line at the symmetric point
 B · the double well V = -mu^2 x^2 + lam x^4: two vacua, one energy; down at the line, up at the vacuum

```
  end do
  call check('1000 wells: V(v) = V(-v) exactly, below V(0); curvature
  -2 mu^2 at the line, +4 mu^2 at the vacuum', ok == 1000)
end subroutine block_b
subroutine block_c()
  real(dp), parameter :: GF = 1.1663788e-5_dp, MH = 125.20_dp, MW =
  80.3692_dp, MZ = 91.1876_dp
  real(dp), parameter :: AINV = 127.951_dp, S2W = 0.23129_dp
  real(dp) :: lam, e, g, mw_p, mz_p
  write(*, '(a)') 'C · the electroweak vacuum: v from the Fermi
  constant, the masses as the depth read by the couplings'
  vev = 1.0_dp/sqrt(sqrt(2.0_dp)*GF)
  lam = MH**2/(2.0_dp*vev**2)
  e = sqrt(4.0_dp*PI/AINV); g = e/sqrt(S2W)
  mw_p = g*vev/2.0_dp; mz_p = e*vev/(2.0_dp*sqrt(S2W)*sqrt(1.0_dp -
  S2W))
  write(*, '(a,f9.4,a,f7.4,a,f8.3,a,f8.3)') ' v = ', vev, ' GeV ;
  lambda = ', lam, ' ; m_W = ', mw_p, ' ; m_Z = ', mz_p
  call check('v = 246.22 GeV from muon decay', abs(vev - 246.22_dp) <
  0.01_dp)
  call check('g v/2 gives m_W within 0.5 percent of 80.369 GeV, and e
  v/(2 s c) m_Z within 0.5 percent of 91.188', &
  abs(mw_p/MW - 1.0_dp) < 5.0e-3_dp .and. abs(mz_p/MZ -
  1.0_dp) < 5.0e-3_dp)
end subroutine block_c
subroutine block_d()
  real(dp), parameter :: MT = 172.57_dp, MB = 4.183_dp, MTAU =
  1.77693_dp, ME = 0.51099895e-3_dp
  real(dp) :: yt, yb, ytau, ye
  write(*, '(a)') 'D · each mass is a coupling to the depth: y =
  sqrt(2) m / v'
  yt = sqrt(2.0_dp)*MT/vev; yb = sqrt(2.0_dp)*MB/vev; ytau =
  sqrt(2.0_dp)*MTAU/vev; ye = sqrt(2.0_dp)*ME/vev
  write(*, '(a,f8.5,a,f8.5,a,f8.5,a,e10.3)') ' top ', yt, ' ; bottom
  ', yb, ' ; tau ', ytau, ' ; electron ', ye
  call check('the top couples at 0.991, the electron at 2.9e-6: 5.5
  decades on one depth', &
  abs(yt - 0.9912_dp) < 1.0e-3_dp .and. abs(log10(yt/ye) -
  5.53_dp) < 0.01_dp)
end subroutine block_d
subroutine block_e()
  integer :: k, it
  real(dp) :: mu2, lam, c, tc, t, x, best, xbest, vt
  logical :: ok_above, ok_below
  write(*, '(a)') 'E · heat restores the line: V + c T^2 x^2 has its
  ground at x = 0 once c T^2 >= mu^2'
  mu2 = 1.0_dp; lam = 0.25_dp; c = 0.5_dp; tc = sqrt(mu2/c)
  ok_above = .true.; ok_below = .true.
  do it = 1, 2
    t = merge(1.2_dp*tc, 0.8_dp*tc, it == 1)
    best = huge(1.0_dp); xbest = 0.0_dp
    do k = -4000, 4000
      x = real(k, dp)*1.0e-3_dp
      vt = pot(mu2, lam, x) + c*t*t*x*x
      if (vt < best) then
        best = vt; xbest = x
      end if
    end do
    write(*, '(a,f6.3,a,f8.4)') ' T/Tc = ', t/tc, ' ; ground at x = ',
    xbest
    if (it == 1) ok_above = (xbest == 0.0_dp)
    if (it == 2) ok_below = (abs(xbest) > 0.5_dp)
  end do
  call check('above Tc the ground is the line; below it the vacuum
  leaves the line', ok_above .and. ok_below)
end subroutine block_e
end program symmetry_breaking_twin
```

PASS 1000 wells: V(v) = V(-v) exactly, below V(0); curvature -2 mu^2 at the line, +4 mu^2 at the vacuum
 C · the electroweak vacuum: v from the Fermi constant, the masses as the depth read by the couplings
 v = 246.2196 GeV ; lambda = 0.1293 ; m_W = 80.223 ; m_Z = 91.499
 PASS v = 246.22 GeV from muon decay

PASS g v/2 gives m_W within 0.5 percent of 80.369 GeV, and e v/(2 s c) m_Z within 0.5 percent of 91.188
D · each mass is a coupling to the depth: $y = \sqrt{2} \cdot m / v$
top 0.99119 ; bottom 0.02403 ; tau 0.01021 ; electron 2.935E-06
PASS the top couples at 0.991, the electron at 2.9e-6: 5.5 decades on one depth

E · heat restores the line: $V + c T^2 x^2$ has its ground at $x = 0$ once $c T^2 \geq \mu^2$
 $T/T_c = 1.200$; ground at $x = 0.0000$
 $T/T_c = 0.800$; ground at $x = -0.8490$
PASS above T_c the ground is the line; below it the vacuum leaves the line
BATTERY-JSON: {"checks":6,"failures":0}

APPENDIX C: THE SEAT AND THE JUDGMENT

PhysOS^T 1.0.5p · RECEIPT · mode quick
TOOLCHAIN Lean (version 4.19.0, x86_64-unknown-linux-gnu, commit 6caaaa842e94, Release) · GNU Fortran (Ubuntu 13.3.0-6ubuntu2-24.04.1) 13.3.0
MANIFEST sha256 a4108cd26bcbe6ef, the files that ran
GROUND toolchain: Lean 4.19.0 and gfortran present · manifest: 31 files signed, all verified · proofs: 2 PhysOS Proofs read from II.10, record lines complete · source screen: 15 kernels, 23 named premises, clean · citations: 253 laws named in Parts I and II, 253 resolved · quarantine: 6 retirements string-scanned, 3 held by review, clean
EXITS ground 0 · registerA RA_T0E_Thesis 0 · registerB Bridge_Final 0 · registerB Universal_Closure 0 · registerB One_Cut 0 · registerB One_Cut_Resolution 0 · registerB One_Cut_Terminal 0 · registerB Only_The_Arrow_Remains 0 · registerB Physical_Closure 0 · registerB Forced_Closure 0 · registerB Closure_Executed 0 · registerB Armed_Seal 0 · registerB Census 0 · registerB Heat_Bridge 0 · registerB The_Loop 0 · registerB RH_Least_Erasure 0 · witness One_Cut_Twin 0 · witness Physical_Closure_Twin 0 · witness Census_Twin 0 · witness Loop_Twin 0 · witness RH_Seal_Twin 0
REGISTER A {"checks":1123,"failures":0,"mode":"sealed"}
WITNESS 5 of 5 twins ran, each held to one battery line with zero failures
CHAIN · D0 83f872725ad9 -> D1 90fc702bbd1 -> D2 492e519daab9 -> D3 6618bec5164b
ELAPSED ground 1 s · Register A 3 s · Register B 34 s · witness 6 s · total 44 s (not hashed)
SEAT EARNED · this run

SCREEN forbidden: none | imports: 0 | axioms declared: 0
laws: 56
JUDGMENT refused 56 of 56 as proof failures | survived none | parse failures none | planted vacuum law survives its negation: True